Summary in Graph

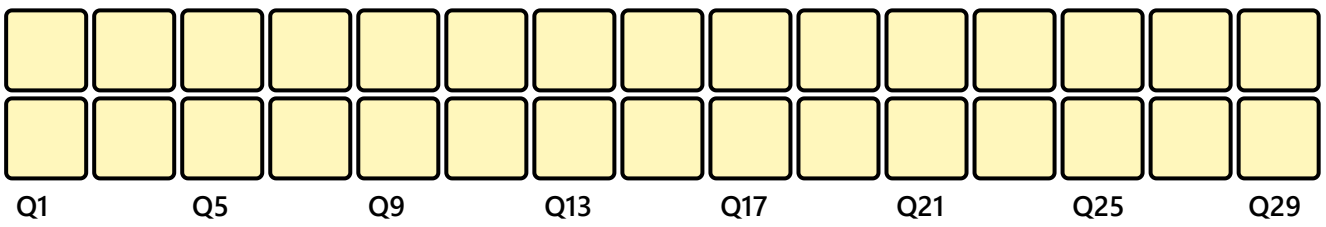
Exam Summary (GO Classes GATE CS/DA | Linear Algebra| Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (20 + 10)
Total Marks:	40 (20 + 20)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (91) out of 905 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ given by

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ t \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 6 \\ t \end{pmatrix}$$

Which statement is true?

A. The vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ are linearly independent for all t
B. The vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ are linearly dependent for all t
C. The vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ are linearly dependent when $t = 4$, and linearly independent otherwise
D. The vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ are linearly independent when $t = 4$, and linearly dependent otherwise

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider a matrix $A_{4\times 5}$., where all the solutions of $Ax = 0$ has the following form -

$$\begin{bmatrix} 6c - 12e \\ -4c + 10e \\ c \\ -5e \\ e \end{bmatrix}$$

What will be the rank of A?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a $n \times n$ matrix, and let u, v, w be nonzero vectors in $\mathbf{R}^n$ which are distinct (so $u \neq v, u \neq w$, and $v \neq w$).
Suppose $Au = 2u, \quad Av = 2v, \quad Aw = -w.$
Which one of the following vectors must be an eigenvector of A?

A. $u - v$
B. $v - w$
C. $u - w$
D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider a matrix $A_{n\times n}$ having the following characteristic equation -

$$\lambda^2(\lambda - 3)(\lambda + 2)^3(\lambda - 4)^3$$

What could be $\mathrm{rank(A)}$?

A. 6
B. 7
C. 8
D. 9

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 03sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the system below, where h and k are real numbers.

$$x + 3y = 2$$
$$3x - hy = k$$

Find the values of h and k which give the system infinitely many solutions.

A. $h = -9$ and $k \neq 6$
B. $h \neq -9$ and k can be any real number
C. $h = -9$ and $k = 6$
D. System is inconsistent for all values of h and k

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $\vec{v}$ be an eigenvector of an invertible matrix A. Which of the following are necessarily true?

A. $\vec{v}$ is an eigenvector of A^{-1}.
B. $\vec{v}$ is an eigenvector of A^2.
C. $\vec{v}$ is an eigenvector of $A + I$.
D. $\vec{v}$ is an eigenvector of $A + 2I$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 3×3 matrix. Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent 3-dimensional vectors. Suppose that we have

$$A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, A\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, A\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Then find the value of the determinant of the matrix A.

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE?

A. If the echelon form of an $m \times n$ matrix has a pivot in every column then $n \geq m$
B. If the echelon form of an $m \times n$ matrix A has a pivot in every row, then the matrix equation $A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \vec{b}$ is consistent for every $\vec{b} \in \mathbb{R}^m$.
C. A system of 25 linear equations in 13 unknowns always has at least one solution.
D. If a system of 25 linear equations in 13 unknowns has at least one solution then it has infinitely many solutions.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

$$\begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} a & 1 & 0 \\ 2 & b & 1 \end{bmatrix} = A_{3\times 3}$$

At what values of (a, b), $A_{3\times 3}$ will be invertible?

A. $(1, -1)$
B. $(-1, 1)$
C. Both A and B
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

The given system has infinitely many solutions for $k =?$

$$\begin{bmatrix} 2 & 2 & -4 \\ 1 & 3 & -2 \\ -4 & k & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$$

Your Answer: | Correct Answer: -4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

In the echelon form of the augmented matrix $[A \mid b]$:

I. A single row of the form $(000 \ldots 0 \mid 0)$ is enough to conclude that the system will have infinitely many solutions.
II. A single row of the form $(00 \cdots 0 \mid 1)$ is enough to conclude that the system will have no solution.

A. Only I is true.
B. Only II is true.
C. Both I and II are true.
D. Neither I, nor II are true.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A, and $\det(\mathrm{A}) = 4$, then α equals to?

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Given the following matrix:

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

Consider the following statements:

I. $A^2 - 4A - 5I = 0$ (where I is identity matrix).
II. $A^{-1} = \frac{(A-4I)}{5}$.

Which of the following options is correct?

A. Only I is true.
B. Only II is true.
C. Both I and II are true.
D. Neither I, nor II are true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Mark all the INCORRECT statements

A. Let A be the matrix of a rotation by angle 30 degree. That is, for any vector x, the angle between x and Ax is always 30 degree. Then A has no real eigenvalues.
B. Let A be an $n \times n$ matrix. Then the eigenvalues of A are the diagonal entries of A.
C. If A is a 3×3 matrix with eigenvalues $\lambda = 1, 2, 3$, then A is singular.
D. If A is a symmetric matrix, then all its eigenvectors are orthogonal.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose $\mathbf{A} = \mathbf{BC}$, where $\mathbf{B}$ is a 4×2 matrix and $\mathbf{C}$ is a 2×4 matrix. Is $\mathbf{A}$ invertible?

A. Yes, $\mathbf{A}$ is invertible.
B. No, $\mathbf{A}$ is not invertible.
C. Depends on $\mathbf{C}$ only.
D. Depends on $\mathbf{B}$ and $\mathbf{C}$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ k & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. For which value of k does the system $A\mathbf{x} = b$ have a unique solution?

A. There is no such value for k.
B. $k = 0$
C. $k = -1$
D. $k = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be an $m \times n$-matrix and let B be an $n \times m$-matrix. Then which of the following statement is not true for all such matrices?

A. BA is defined
B. the columns of AB are linear combinations of the columns of B
C. AB is defined
D. the columns of AB are linear combinations of the columns of A

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix}$. Let $\boldsymbol{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Which of the following statements is true?

A. $\boldsymbol{v}$ is a linear combination of the columns of B but not of the columns of A.
B. $\boldsymbol{v}$ is a linear combination of the columns of A and of the columns of B.
C. $\boldsymbol{v}$ is a linear combination of the columns of A but not of the columns of B.
D. $\boldsymbol{v}$ is neither a linear combination of the columns of A nor of the columns of B.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a 2×3-matrix and b a vector in $\mathbb{R}^2$.

Consider the following two statements:

- $(\mathrm{P1})$ A has at most two pivots,
- $(\mathrm{P2})$ Assuming $Ax = b$ has a solution, then it has infinitely many solutions.

Then:

A. Only Statement $\mathrm{P2}$ is correct.
B. Neither Statement $\mathrm{P1}$ nor Statement $\mathrm{P2}$ is correct.
C. Only Statement $\mathrm{P1}$ is correct.
D. Statement $\mathrm{P1}$ and Statement $\mathrm{P2}$ are correct.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose A is 3 by 4, and $Ax = 0$ has all solutions in the following form -

$$\mathrm{x} = \mathrm{s}\begin{bmatrix}1\\1\\1\\0\end{bmatrix} + \mathrm{t}\begin{bmatrix}-2\\-1\\0\\1\end{bmatrix}$$

We do some unknown elementary row transformations on A and get the following matrix R -

$$R = \begin{bmatrix}1 & 0 & \mathrm{a} & \mathrm{b}\\0 & 1 & \mathrm{c} & \mathrm{d}\\0 & 0 & 0 & 0\end{bmatrix}$$

What will be the values of $a, b, c,$ and d?

A. $a = -1$
B. $b = 2$
C. $c = -1$
D. $d = 1$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider a matrix A of dimension $m \times n$ such that -

$Ax = \begin{bmatrix}1\\1\\1\end{bmatrix}$ has no solutions and $Ax = \begin{bmatrix}0\\1\\0\end{bmatrix}$ has exactly one solution

Which of the following CAN be true?

A. $\mathrm{Rank}(A) = 2$
B. $m = 3$
C. $n = 1$
D. $\mathrm{Rank}(A) = 1$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

A three-by-three matrix B is known to have eigenvalues $0, 1$ and 2. This information is enough to find which one of these (give the answers where possible):

A. The rank of B
B. The determinant of $B^T B$
C. The eigenvalues of $B^T B$
D. The eigenvalues of $\left(B^2 + I\right)^{-1}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$-matrix. Consider the system of linear equations $AX = b$, which of the following statement is always true:

A. Suppose $m > n$ then rank of augmented matrix $(A \mid b)$ can not larger than m.
B. Suppose A has full rank then $AX = b$ always has a solution.
C. If $AX = b$ has a solution then $AX = 0$ has unique solution.
D. If $AX = b$ has a unique solution then A has to be invertible.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a 3×3 matrix. Suppose that A has eigenvalues 2 and -1, and suppose that $\mathbf{u}$ and $\mathbf{v}$ are eigenvectors corresponding to 2 and -1, respectively, where

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Then compute $A^5\mathbf{w}$, where

$$\mathbf{w} = \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix}$$

A. $\begin{bmatrix} 92 \\ 2 \\ -96 \end{bmatrix}$

B. $\begin{bmatrix} -92 \\ -2 \\ -96 \end{bmatrix}$

C. $\begin{bmatrix} 92 \\ -2 \\ -96 \end{bmatrix}$

D. $\begin{bmatrix} 92 \\ -2 \\ 96 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $u \in \mathbb{R}^n$ be such that $u^T u = 1$ and set $A = uu^T$. What will be the sum of all eigenvalues of A?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Given a matrix $A\,(m \times n)$ and $Ax = b$. Consider below statements :

- $\mathrm{S} : m < n$
- $\mathrm{P}:\ A$ has m Linearly Independent columns
- R : There is a solution for every b in R^m.

Which of the following logical implications are ALWAYS true?

A. $\mathrm{S} \rightarrow \mathrm{R}$
B. $\mathrm{R} \rightarrow \mathrm{P}$
C. $\mathrm{S} \rightarrow \sim \mathrm{P}$
D. $(S$ and $\mathrm{P}) \rightarrow \mathrm{R}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following set of (column) vectors:

$$X = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \mid 2x_1 + 3x_2 - x_3 = 0 \right\}.$$

Which of the following statements are true?

I. Every element of set X can be written as linear combination of vectors of $S = \left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$

II. $\left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$ is a linearly independent set.

Choose the correct option.

A. Only I is true.
B. Only II is true.
C. Both I and II are true.
D. Neither I, nor II are true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Choose the correct statements.

A. If $b > \frac{2}{3}$, then $A = \begin{bmatrix} 2 & b \\ 3 & 1 \end{bmatrix}$ is invertible.
B. If P is an invertible matrix such that $B = P^{-1}AP$, then $\det A = \det B$.
C. Let A be an $m \times n$ matrix. If $A = PDP^{-1}$ for some invertible matrix P, and diagonal matrix D, then for any integer $k \geq 1$, $A^k = PD^kP^{-1}$.
D. A system of linear equations with m equations and n variables, such that $m = n$, always has a solution.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given matrix $A\ (m \times n)$ such that $Ax = b$. Consider below statements :

- $\mathrm{S1}$: If b is Linearly Dependent on cols of matrix, then it need not always have a unique solution.
- $\mathrm{S2}$: If columns of matrix are Linearly Dependent, then above system of equation can have a unique solution.

A. $\mathrm{S1}$ and $\mathrm{S2}$ are true
B. $\mathrm{S1}$ and $\mathrm{S2}$ are false
C. $\mathrm{S1}$ is true and $\mathrm{S2}$ is false
D. $\mathrm{S2}$ is true and $\mathrm{S1}$ is false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

There exist a 3×3 real symmetric matrix S such that -

- $\text{Statement } 1 : \mathrm{S}\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}2\\4\\6\end{pmatrix}$ and $S\begin{pmatrix}2\\1\\0\end{pmatrix} = \begin{pmatrix}-2\\-1\\0\end{pmatrix}$.
- $\text{Statement2} : \mathrm{S}^2 = -\mathrm{I}$

Which of the following option is CORRECT?

A. Statement 1 is true but Statement 2 is false
B. Statement 1 is false but Statement 2 is true
C. Both Statements are true
D. Both Statements are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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